

Name: _____ Date: _____

Period: _____

Primary Objective**I will be able to...**

1. **LO 4.6.A** — Write the standard form equation of a parabola, ellipse (including circles), or hyperbola given key features; and identify those features from a given equation. _____

Vocabulary & Key Concepts

conic section: A curve formed by the intersection of a plane and a double cone: a _____, an _____, or a _____.

vertex (parabola): The point (h, k) in the standard form $y - k = a(x - h)^2$ or $x - h = a(y - k)^2$. It is the _____ point of the parabola.

center (ellipse / hyperbola): The point (h, k) at the _____ of an ellipse or hyperbola; read from h and k in the standard form.

semi-major / semi-minor axis: For an ellipse: the larger radius is a (_____) and the smaller is b (_____).

asymptote (hyperbola): Lines the hyperbola approaches but never reaches: $y - k = \pm(x - h)$. Fill in b over a .

Hook

Entry Question: Planetary orbits, telescope mirrors, and navigation systems all use conic sections.

Q1. Name one real-world application of a parabola and one of an ellipse. _____

Q2. What do you predict all three conic shapes have in common algebraically? _____

Section 1 — Parabolas (LO 4.6.A, EK 4.6.A.1)

Standard forms:

Form	Axis	Direction
$y - k = a(x - h)^2$	vertical	$a > 0$: opens _____; $a < 0$: opens _____
$x - h = a(y - k)^2$	horizontal	$a > 0$: opens _____; $a < 0$: opens _____

Worked Example: $y + 3 = 2(x - 1)^2$

Rewrite: $y - (_) = 2(x - _)^2$

Vertex: _____ $a =$ _____, so opens: _____

Student Practice: $y - 5 = -(x + 2)^2$

Vertex: _____ $a =$ _____, so opens: _____

Section 2 — Ellipses and Circles (LO 4.6.A, EK 4.6.A.2)

Standard form of an ellipse:

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$$

Worked Example: $\frac{(x + 1)^2}{16} + \frac{(y - 3)^2}{9} = 1$

Center: _____ $a^2 =$ _____ $\Rightarrow a =$ _____ (horizontal) $b^2 =$ _____ $\Rightarrow b =$ _____
(vertical)

Circle (special case): When $a = b = r$, the equation becomes

$$(x - h)^2 + (y - k)^2 = r^2$$

Student Practice: $\frac{(x - 3)^2}{25} + \frac{(y + 1)^2}{4} = 1$

Center: _____ $a =$ _____ $b =$ _____

Section 3 — Hyperbolas (LO 4.6.A, EK 4.6.A.3)

Standard forms:

Form

Opens

$$\frac{(x - h)^2}{a^2} - \frac{(y - k)^2}{b^2} = 1$$

left and right (x -term positive)

$$\frac{(y - k)^2}{b^2} - \frac{(x - h)^2}{a^2} = 1$$

up and down (y -term positive)

Asymptotes: $y - k = \pm \frac{b}{a}(x - h)$

Worked Example: $\frac{(x - 2)^2}{9} - \frac{(y + 1)^2}{4} = 1$

Center: _____ $a =$ _____ $b =$ _____ Opens: _____

Asymptotes: $y + 1 = \pm$ _____ $(x - 2)$

Student Practice: $\frac{(y + 3)^2}{16} - \frac{(x - 1)^2}{9} = 1$

Center: _____ $a =$ _____ $b =$ _____ Opens: _____

Asymptotes: $y + 3 = \pm$ _____ $(x - 1)$